

Knowledge Check 2

FreshBox: Launching a Premium Subscription Service

8 questions • 20 points

Business scenario

FreshBox is a national grocery delivery company considering launching a premium subscription service. For a fixed monthly fee, subscribers would receive unlimited free deliveries.

The company wants to estimate demand for the new service and ensure that its delivery operations meet customer expectations. You are a business analyst working with FreshBox's management team. Your task is to help the company design its customer survey and evaluate its delivery performance.

Part 1. Understanding customer preferences

FreshBox has 100,000 existing customers across the United States. The company wants to understand their willingness to pay for the new subscription.

Question 1. [2 points]

Management believes that customers who order groceries frequently may have different preferences from customers who order only occasionally. You want to survey 2,000 customers while ensuring that both groups are adequately represented.

Which sampling strategy would you recommend?

- A. Simple random sampling
- B. Stratified random sampling
- C. Cluster sampling

Solution: B. Stratified random sampling. Divide customers into frequent and occasional shoppers, then randomly sample customers within each group. This ensures sufficient representation of both groups.

Question 2. [2 points]

To reduce costs, FreshBox considers randomly selecting 10 cities and surveying 200 existing customers in each selected city. What is the main statistical drawback of this approach compared with a simple random sample of 2,000 customers nationwide?

- A. Customers in the same city may have similar preferences, reducing the effective amount of information collected.
- B. The sample is necessarily biased because only 10 cities are selected.
- C. Every customer must have an equal probability of being selected for the sample to be valid.
- D. Cluster sampling cannot be used to estimate average willingness to pay.

Solution: A. Customers within a city may have correlated preferences. This can reduce the effective amount of independent information relative to a simple random sample of the same size.

Question 3. [4 points]

FreshBox decides to send its survey only to existing customers in San Francisco. Participation is voluntary.

Complete each statement using the appropriate statistical concept: *sampling frame bias*, *voluntary response bias*, *simple random sampling*, *cluster sampling*.

- Customers in San Francisco may have different preferences from customers elsewhere in the United States. This is an example of: _____. **Sampling frame bias.**
- Customers with particularly strong preferences for the subscription may be more likely to complete the survey. This is an example of: _____. **Voluntary response bias.**

Question 4. Sampling weights [2 points]

FreshBox decides to use stratified random sampling to ensure that both frequent and occasional customers are adequately represented. The company randomly selects 500 customers from each group and obtains the following results:

Customer group	Share of all customers	Average willingness to pay
Frequent customers	20%	\$30/month
Occasional customers	80%	\$15/month

What is the estimated average willingness to pay for the subscription across all FreshBox customers? Enter your answer in dollars, rounded to two decimal places.

Answer: \$_____

Solution: Use population shares, not the equal sample shares, as weights:

$$\bar{X}_w = 0.20(30) + 0.80(15) = \boxed{\$18.00}.$$

The unweighted average, \$22.50, would overrepresent frequent customers.

Question 5. [2 points]

FreshBox wants to estimate the average amount customers spend per order. The population standard deviation of order values is \$40. Management wants the standard error of the sample mean to be no greater than \$2. What is the minimum number of independent customer orders FreshBox must randomly sample to achieve this goal?

Answer: _____

Solution: 400 orders Using the standard error formula:

$$\frac{40}{\sqrt{n}} \leq 2$$

Therefore,

$$\sqrt{n} \geq 20 \implies n \geq 400.$$

Part 2 – Monitoring delivery performance

FreshBox has set a target mean delivery time of 30 minutes for its premium subscribers. The company wants to monitor whether its delivery operations are meeting this target. Assume that delivery times are independent and normally distributed, with a known population standard deviation of 8 minutes. Every morning, the operations team randomly selects 64 deliveries and calculates their average delivery time.

The company uses a two-sided control procedure: if the sample mean falls outside the control limits, management investigates and adjusts the delivery process. Both unusually long and unusually short delivery times are considered deviations from the target.

Question 6.

[2 points]

Complete the following statements using the terms *Type I error* and *Type II error*.

- FreshBox concludes that the delivery process is off target and makes unnecessary operational adjustments, even though the true mean delivery time is 30 minutes. This is a: _____.
Type I error.
- FreshBox concludes that the delivery process is on target and makes no adjustments, even though the true mean delivery time differs from 30 minutes. This is a: _____. **Type II error.**

Question 7.

[4 points]

FreshBox uses a two-sided control procedure with a 5% probability of a Type I error. The company is considering changing its sample size from 64 to 100 deliveries per day, while keeping the Type I error rate unchanged.

For each statement, select **True** or **False**.

- Increasing the sample size will reduce the standard error of the sample mean.

☐ True ☐ False

True. The standard error is $SE(\bar{X}) = \frac{\sigma}{\sqrt{n}}$. Holding the population standard deviation fixed, increasing n increases the denominator and therefore reduces the standard error.

- Increasing the sample size will make the control limits narrower.

☐ True ☐ False

True. For a fixed Type I error rate, the control limits are determined by the target mean plus or minus a fixed multiple of the standard error. Since a larger sample reduces the standard error, the control limits move closer to the target and become narrower.

- c. Increasing the sample size will increase the probability of a Type I error.

☐ True ☐ False

False. The Type I error rate remains fixed at 5%.

- d. Increasing the sample size will make it easier to detect a genuine deviation from the 30-minute target.

☐ True ☐ False

True. A larger sample size reduces the standard error, so the sampling distribution of the sample mean is more concentrated. This makes it more likely that a genuine deviation from the 30-minute target will be detected. In other words, increasing the sample size increases the **power** of the procedure.

Question 8.

[2 points]

Suppose FreshBox's control limits are **28.5 minutes** and **31.5 minutes**. On a particular day, the average delivery time in the sample is **32.2 minutes**.

- a. Which of the following is the most appropriate conclusion?

- ☐ The true mean delivery time is definitely greater than 30 minutes.
☐ There is evidence that the delivery process may be off target.
☐ A Type I error must have occurred.
☐ The delivery process is on target because 32.2 minutes is close to 30 minutes.

Answer: There is evidence that the delivery process may be off target. The sample mean of 32.2 minutes is above the upper control limit of 31.5 minutes, so the procedure signals a potential deviation from the target.

- b. FreshBox investigates but discovers that the true mean delivery time is actually 30 minutes. In this case, the original signal from the control procedure was:

- ☐ A Type I error
☐ A Type II error
☐ A correct rejection
☐ A correct non-rejection

Answer: Type I error. FreshBox concluded that the process was off target even though the true mean was 30 minutes.